Discrete Systems: The Dripping Faucet

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Physics 4428 Laboratory Fall 2026

Experimental Overview

The dripping faucet was studied as an event-driven nonlinear system in which the time between successive drops depends on flow conditions and on the state left by previous drops. A reservoir, valve, and pipette supplied the drops; a photogate recorded successive beam-blocking times. The measured quantity was the blocked-to-blocked interval Δt_n, and acquisition time located each interval in a run. Five recorded regimes were compared using interval-versus-time plots, return maps Δt_{n+1} versus Δt_n, and interval histograms. These views distinguished nearly constant, fluctuating, alternating, approximately three-step, and irregular sequences; no flow-rate measurements or perturbation tests were available to establish a full bifurcation diagram or mathematical chaos.

Data, Analysis, and Notes

For each CSV file, rows with a numeric `Latest: DeltaT` value were retained in their recorded order. Empty photogate rows and Logger Pro histogram columns were excluded. The time plot preserves temporal order, the return map exposes one-step dependence, and the histogram shows how often each interval occurs but discards ordering. The sample standard deviation s and coefficient of variation $s/\overline{\Delta t}$ summarize spread. Each run lasted about 20 seconds at one unrecorded valve setting, so run-to-run comparisons are qualitative rather than a calibrated flow-rate sweep.

Regime	N	Mean (s)	s (s)	Range (s)	CSV file
Periodic	44	0.4456	0.0020	0.4329–0.4464	`Drip_PwithnoDeviation.csv`
Fluctuating	72	0.2734	0.0032	0.2652–0.2771	`Drip_PwithDeviation.csv`
Period two	51	0.3916	0.0030	0.3876–0.3969	`Drip_Period2_3.csv`
Irregular	272	0.0732	0.0127	0.0454–0.0914	`Drip_Chaos.csv`
Other: three-step	84	0.2350	0.0043	0.2221–0.2402	`Drip_Period2.csv`

Table 1: Summary of the five runs used for the required behavior comparison. The labels describe the observed sequences; the filenames are acquisition labels.

The solid curve on each return map is a least-squares quadratic of the form $\Delta t_{n+1} = a(\Delta t_n - \overline{\Delta t_n})^2 + b(\Delta t_n - \overline{\Delta t_n}) + c$, with intervals expressed in milliseconds. The dashed line is $\Delta t_{n+1} = \Delta t_n$. These empirical fits describe the recorded points; they are not an established equation of motion for the faucet.

Periodic behavior

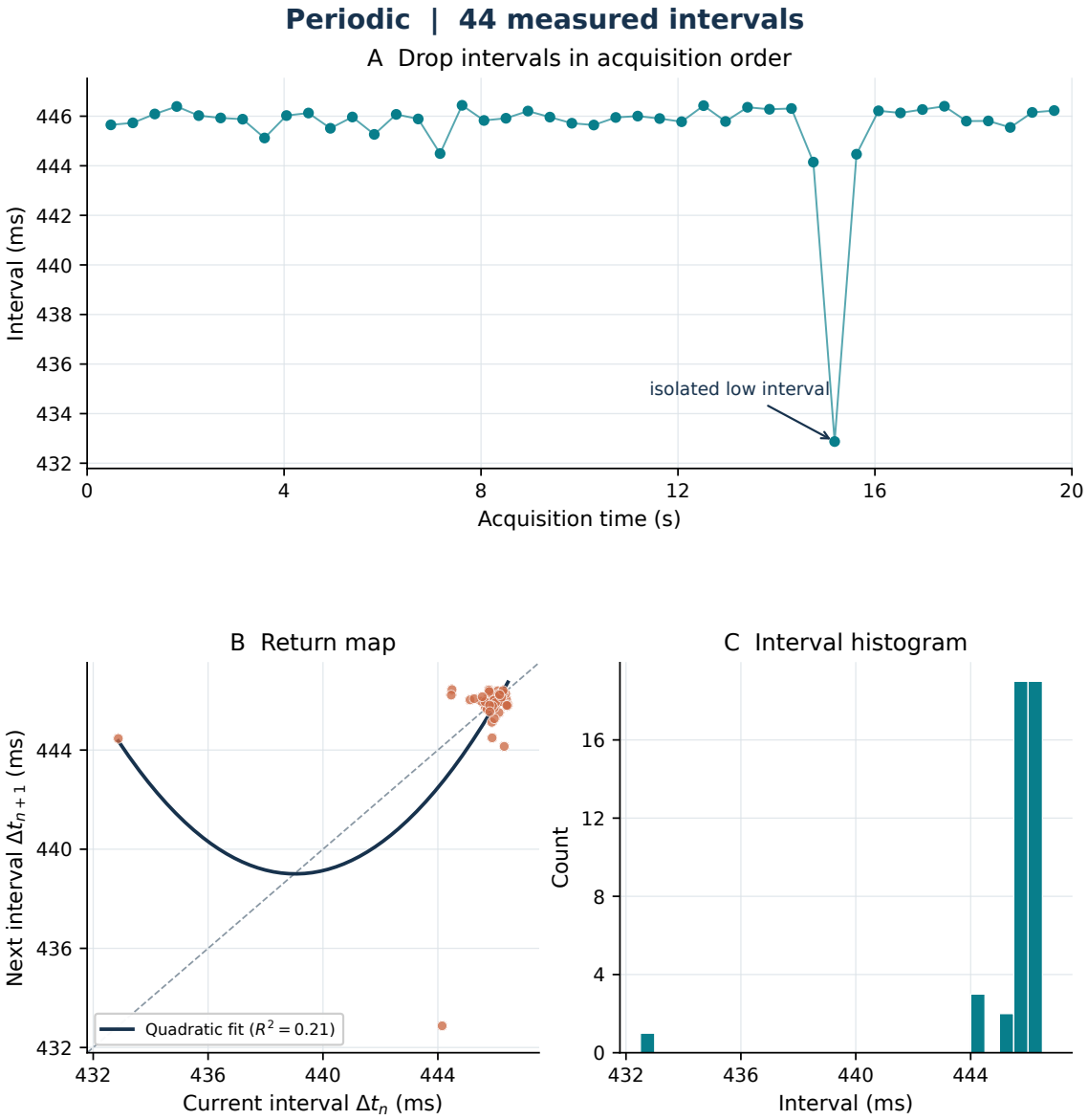


Figure 1: Nearly constant drop intervals: time series, return map, and histogram.

Notes. Most intervals cluster near 0.446 s, producing a nearly horizontal time trace, a compact return-map cluster near $(0.446, 0.446)$, and one narrow histogram peak. One early interval near 0.433 s increases the reported spread. The compact cluster is consistent with a stable one-interval state over this run.

Periodic behavior with fluctuations

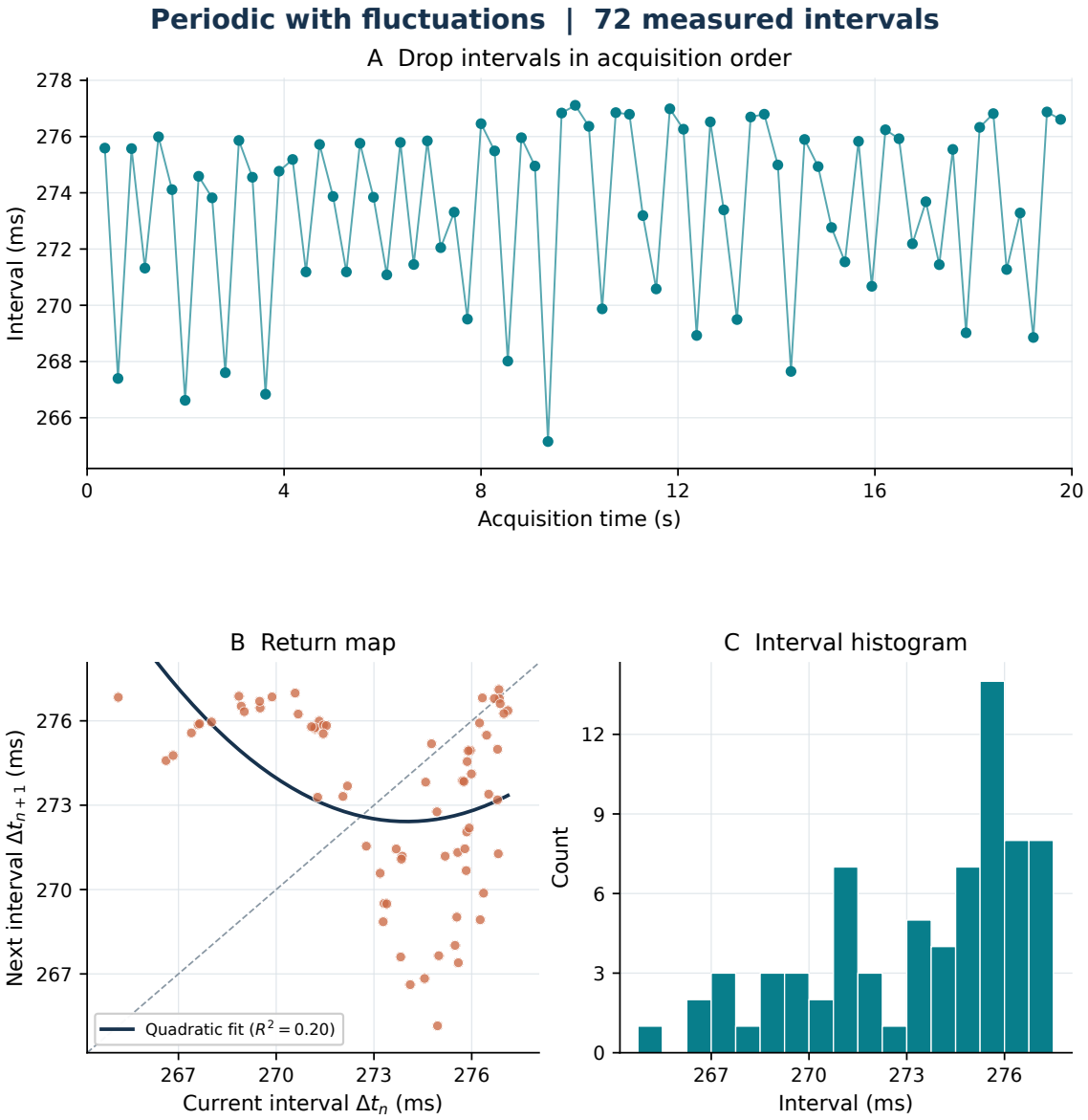


Figure 2: Small deviations around the repeated drop interval.

Notes. The time trace remains near 0.273 s but is visibly less steady than the periodic run. The standard deviation is about 1.2% of the mean, while the full measured excursion is about 4.4% of the mean; this is smaller than the handout's suggested 5–10% fluctuation scale. A broadened and structured histogram and a spread return map show variations in interval size. The data do not establish whether these changes came from weak deterministic dynamics, slow drift, or apparatus noise.

Period-two behavior

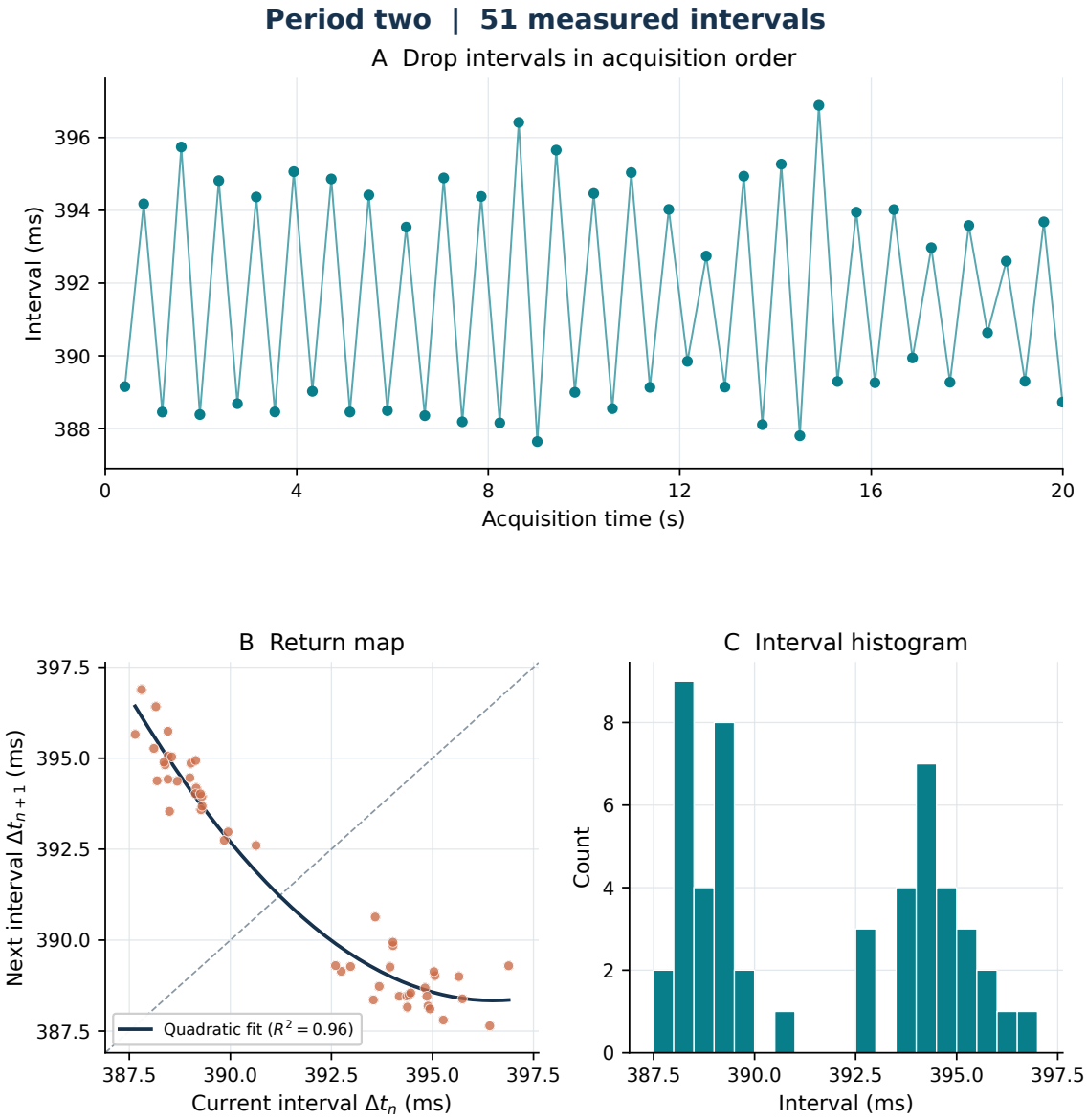


Figure 3: Alternating intervals in `Drip_Period2_3.csv`.

Notes. The intervals alternate between groups centered near 0.389 s and 0.395 s. The return map has two off-diagonal clusters: a short interval is followed by a long one, then by a short one. Two histogram peaks confirm two favored values, but the alternation is established by the time sequence and return map. The lag-one correlation is -0.963 and the lag-two correlation is 0.944, supporting an approximate two-step cycle.

Irregular behavior

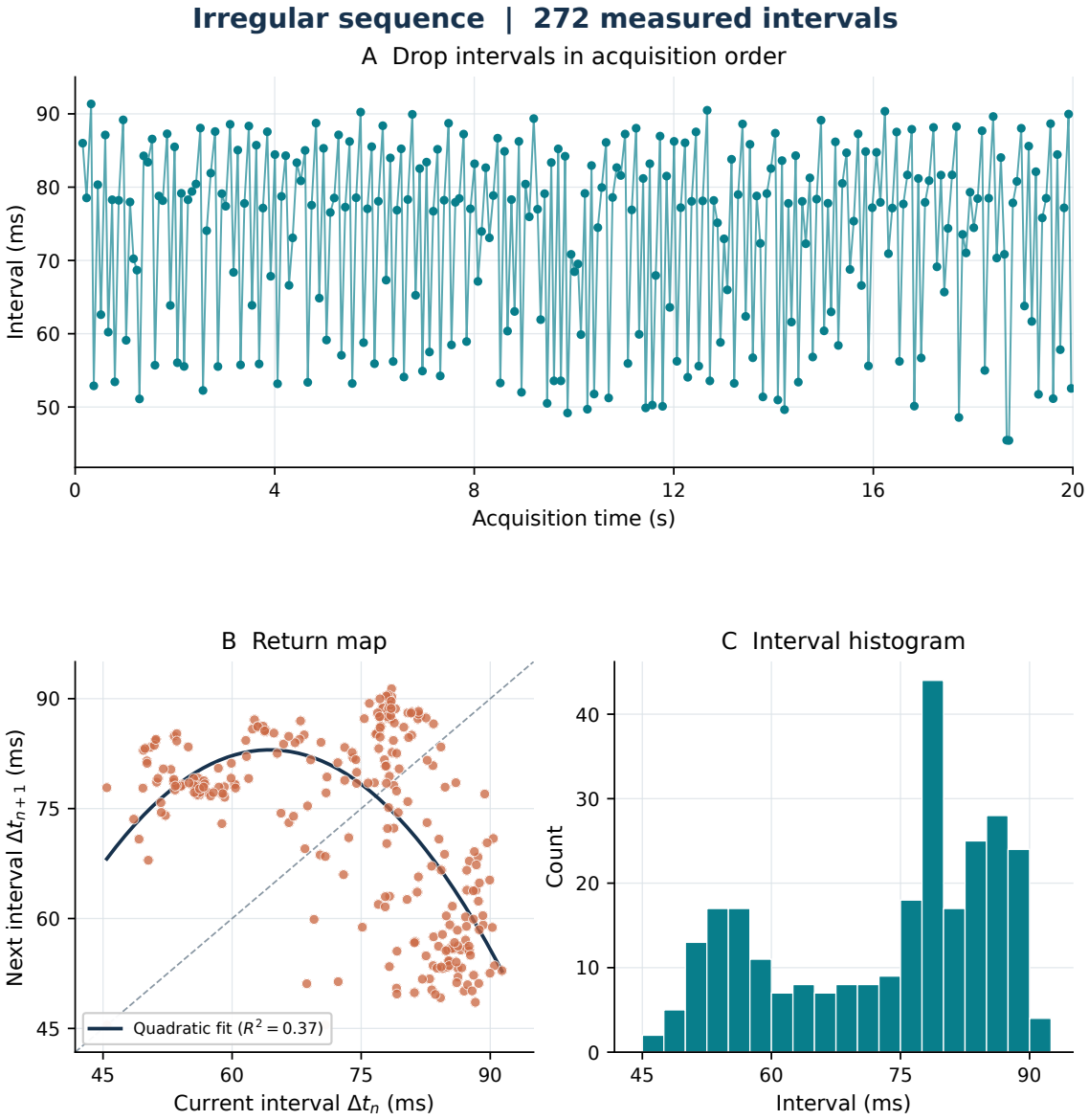


Figure 4: Broad, irregular interval sequence from `Drip_Chaos.csv`.

Notes. The time series lacks a persistent one-, two-, or three-step repeat. Its histogram spans 0.0454 s to 0.0914 s, and its return map occupies a much broader set than the periodic runs. The coefficient of variation is 17.4%. This is an irregular or chaotic *candidate*, not a proof of deterministic chaos: the short record, unknown measurement noise, and possible missed drops limit that conclusion. Seven of the 272 measured intervals are at or below 0.050 s, where the handout warns that the counter can miss events.

Other pattern: approximate period three

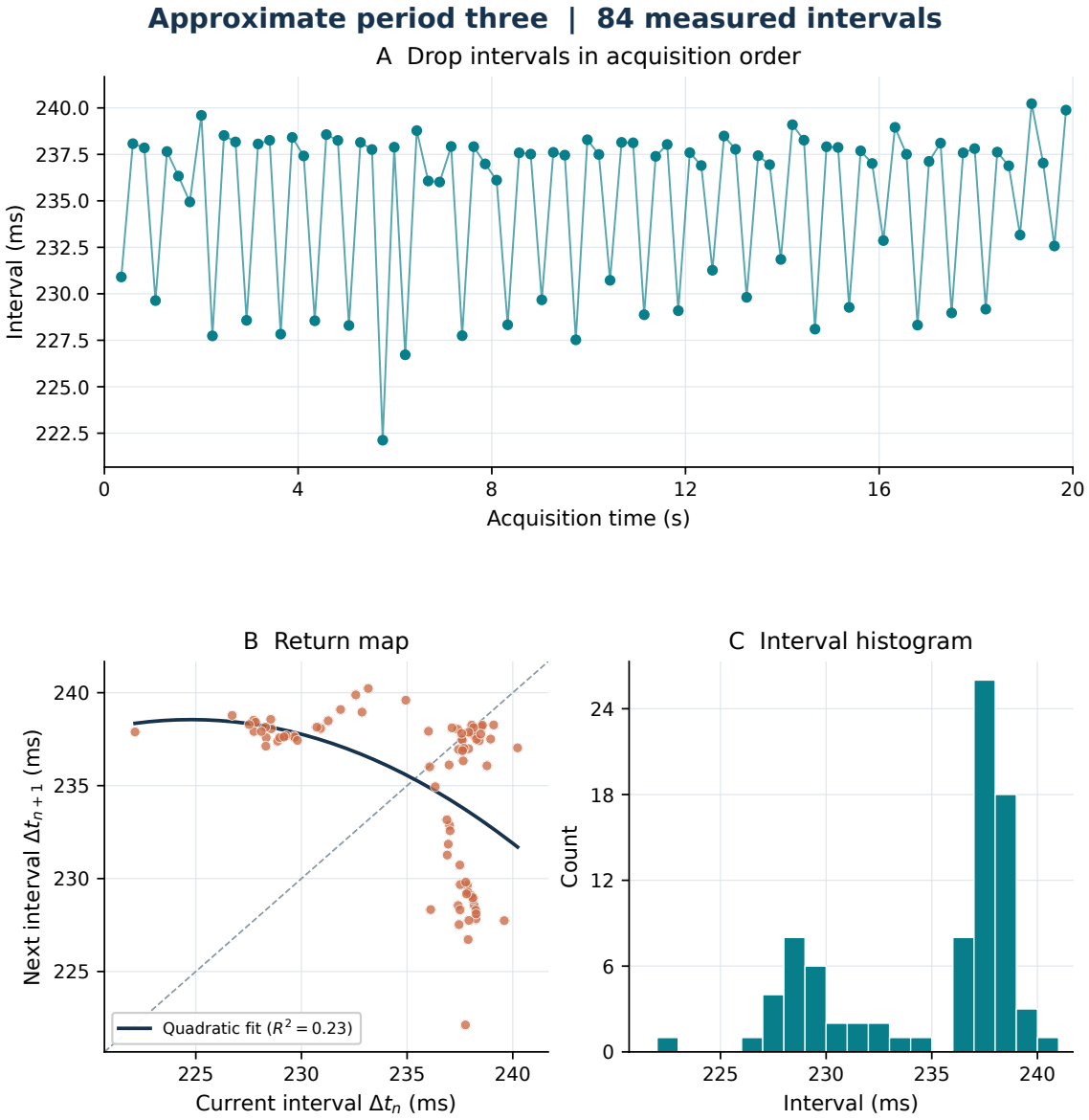


Figure 5: An approximate three-step pattern in `Drip_Period2.csv`.

Notes. Despite the filename, this run often has a short interval near 0.229 s followed by two longer intervals near 0.238 s. The return map has more structure than the two-point alternating map, and the histogram has a low group plus a higher group; the two long steps overlap in the histogram. The lag-three correlation is 0.659, greater than its lag-one and lag-two correlations (−0.468 and −0.279). Thus "approximate period three" describes the observed recurrence, while departures from an exact cycle remain visible.

Answers to Questions

1. **How this system differs from the earlier systems.** The measured state is a sequence of event-to-event intervals, rather than a continuously sampled position or amplitude. Each drop detaches, resets the meniscus, and affects the next interval. The dynamics are therefore naturally represented by a discrete return map, even though fluid motion between drops is continuous.

2. **Histogram signatures and period four.** One narrow peak indicates a single favored interval; two separated peaks are consistent with period two. More peaks indicate more favored intervals, while a wide or irregular distribution signals greater variation. A histogram cannot establish the order of the intervals: a genuine period-four sequence also requires the time plot or return map to show a four-step repeat. Figure 6 sketches the ideal four-peak histogram; unequal peak heights or merged peaks are possible in measured data.

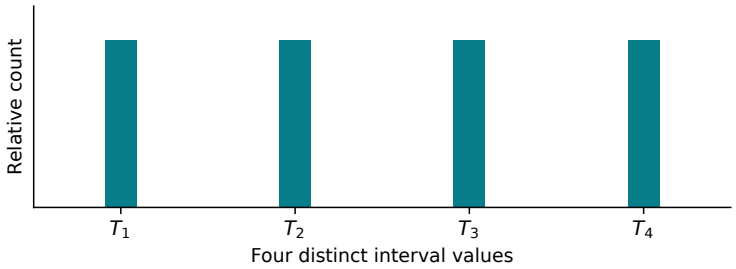


Figure 6: Idealized histogram for four distinct, equally visited drop intervals. This is a schematic, not measured data.

3. **Logistic-map comparison for the 50% soap/50% water fluid.** In the logistic map $x_{n+1} = rx_n(1-x_n)$, a fixed point corresponds to one repeated interval, a two-cycle to alternating intervals, and a chaotic regime to bounded, nonrepeating intervals. The observed one- and two-cluster return maps illustrate the first two patterns. The broad irregular run resembles the third qualitatively, but these CSV files do not identify a 50% soap/50% water mixture, measure r, or establish that the faucet follows the logistic equation. No mixture-specific transition is inferred from these data.

4. **Control parameter along the map's horizontal axis.** In a logistic-map bifurcation diagram the horizontal control parameter is r, not elapsed time. For the physical faucet the analogous adjustable quantity is flow rate through the pipette, set by valve opening and pressure head. A changing setting during acquisition would make it vary with time, but the supplied CSV files record neither valve position nor flow rate. The plotted time axes here show acquisition time within individual runs.

Limitations and Conclusion

The five runs show a progression of distinct interval patterns, including a stable single interval, weak fluctuations, alternating intervals, a three-step recurrence, and a broad irregular sequence. Since actual flow rates were not recorded, their order cannot be used as a quantitative bifurcation diagram. Photogate misses near 0.05 s and the absence of a noise or sensitivity analysis also prevent a rigorous chaos claim. The return maps, interpreted together with the time sequences and histograms, provide the strongest evidence for the periodicity classifications.

Appendix A: Analysis Code

The Python code reads the numeric drop intervals from the supplied CSV files and generates the plots in this report. The plotting script in the zip includes the figure styling and period-four schematic.

```
from pathlib import Path
import csv
import matplotlib.pyplot as plt
import numpy as np

root = Path(__file__).resolve().parent
runs = [
    ("periodic", "Drip_PwithnoDeviation.csv", 0.5),
    ("fluctuating", "Drip_PwithDeviation.csv", 0.75),
    ("period_two", "Drip_Period2_3.csv", 0.5),
    ("irregular", "Drip_Chaos.csv", 2.5),
    ("period_three", "Drip_Period2.csv", 1.0),
]

for name, filename, bin_width in runs:
    with (root / "data" / filename).open(newline="") as file:
        events = [(float(r["Latest: Time (s)"]),
                   float(r["Latest: DeltaT"]))
                  for r in csv.DictReader(file) if r["Latest: DeltaT"]]
    time, seconds = np.asarray(events).T
    interval = seconds * 1000  # seconds to milliseconds

    fig = plt.figure(figsize=(7.2, 7.4), layout="constrained")
    grid = fig.add_gridspec(2, 2)
    time_ax = fig.add_subplot(grid[0, :])
    map_ax = fig.add_subplot(grid[1, 0])
    hist_ax = fig.add_subplot(grid[1, 1])
    time_ax.plot(time, interval, "o-", markersize=3)
    map_ax.scatter(interval[:-1], interval[1:], s=18)
    x, y = interval[:-1], interval[1:]
    center = x.mean()
    a, b, c = np.polyfit(x - center, y, 2)
    curve_x = np.linspace(x.min(), x.max(), 300)
    curve_y = a * (curve_x - center)**2 + b * (curve_x - center) + c
    map_ax.plot(curve_x, curve_y, label="Quadratic fit")
    edges = np.arange(interval.min(), interval.max() +
                      2 * bin_width, bin_width)
    hist_ax.hist(interval, bins=edges)
    time_ax.set(xlabel="Time (s)", ylabel="Interval (ms)")
    map_ax.set(xlabel="Current interval (ms)",
               ylabel="Next interval (ms)")
    hist_ax.set(xlabel="Interval (ms)", ylabel="Count")
    fig.savefig(root / "figures" / f"{name}.pdf")
    plt.close(fig)
```

Sources

Physics 4428, *Discrete Systems: The Dripping Faucet*, Fall 2026 lab handout.
Physics 4428, *Laboratory Syllabus*, Fall 2026, lab report format.